UNIT 5 · TOPIC 5(B)

Machine-Specific Hazards and Guarding

Syllabus topic: Safety aspects of agricultural machinery

PTO entanglement

- Loose clothing, dupatta or hair caught in an unguarded shaft causes severe degloving/amputation and death.
- **Mandatory:** master shield at the tractor + a freely-rotating tunnel guard on the shaft.
- Never step over a rotating shaft; disengage the PTO before dismounting.

Threshers — the biggest Indian machinery-injury category

Hands and arms are drawn into the cylinder while feeding. Controls:

- **Safe feeding-chute geometry (IS 9129)** — a long/angled chute so the hand cannot reach the cylinder.
- Feeding-roller types; emergency stop; guarding of belts and pulleys (**IS 9020** safety requirements; **IS 6284** test code — Unit 3, Topic 4b).
- Never feed in loose clothing, never work fatigued at night, keep children away.

BIS has withdrawn IS 9129 and folded it into IS 9020 (Power Threshers — Safety Requirements, revised 2002). Exam keys may cite either.

Other hazards

Hazard	Control
Belt/pulley and chain drives (nip points)	Guards
Rotavator blades	Stop the PTO before inspection
Harvester augers, combine headers	Guards; stop before clearing
Hydraulic high-pressure pinhole leaks	Never search for leaks with bare hands (injection injury)
Stored energy in raised implements	Block/support mechanically before working underneath
Battery	Hydrogen explosion and acid burns
Fuel	Fire
Silo / manure pits	Toxic gases
Highway transport	SMV emblem, lights, brake pedals latched together, safe towing limits

Safety design features to know

- Shields and interlocks; guarded PTO; hydraulic lock valves; dead-man controls.
- **Neutral-start interlock:** the engine can start only in neutral with the PTO off.
- Audible reverse alarms.
- ISO/ASAE universal control SYMBOLS and colour coding (ASAE EP443 — Liljedahl p. 230) so controls are recognised across makes.

- Instruments grouped in the operator's 30° visual cone (p. 229; Topic 10b).

Practice MCQs

Q85. The largest single cause of machinery injuries in Indian farms is associated with:

(a) Seed drills
(b) Threshers (cylinder entanglement while feeding)
(c) Ploughs
(d) Levellers

Q86. The Indian standard prescribing safe thresher feeding-chute geometry is:

(a) IS 9129
(b) IS 6316
(c) IS 4468
(d) IS 2062

Q87. Safety requirements for power threshers are in:

(a) IS 9020
(b) ISO 500
(c) IS 1239
(d) SAE J709

Q88. PTO shaft safety requires:

(a) Only paint
(b) Master shield plus freely-rotating tunnel guard
(c) A chain
(d) Nothing

Q89. Loose clothing near a rotating PTO shaft chiefly risks:

(a) Cooling
(b) Entanglement and severe limb injury
(c) Corrosion
(d) Noise

Q90. A hydraulic pinhole leak should NEVER be located with:

(a) Cardboard
(b) Bare hands (injection injury risk)
(c) Goggles on
(d) Engine off

Q91. Before working under a raised implement one must:

(a) Trust the hydraulics
(b) Block/support it mechanically
(c) Increase pressure
(d) Idle the engine

Q92. A neutral-start interlock ensures the engine starts only when:

(a) PTO is engaged
(b) Transmission is in neutral/PTO off
(c) Lights are on
(d) Fuel is full

Answers

Q85 — (b) Threshers (cylinder entanglement while feeding)
Q86 — (a) IS 9129
Q87 — (a) IS 9020
Q88 — (b) Master shield plus freely-rotating tunnel guard
Q89 — (b) Entanglement and severe limb injury
Q90 — (b) Bare hands (injection injury risk)
Q91 — (b) Block/support it mechanically
Q92 — (b) Transmission is in neutral/PTO off

Sources: Liljedahl et al. — Tractors and their Power Units, Ch. 9, pp. 231–235; ASAE S383.1 & S310.3; BIS safety standards IS 9020 / IS 9129 / IS 12239; Insecticides Act 1968 & Rules 1971 (label classes); WHO hazard classification; ICAR-CIAE / AICRP farm-accident surveys; NPTEL lectures.